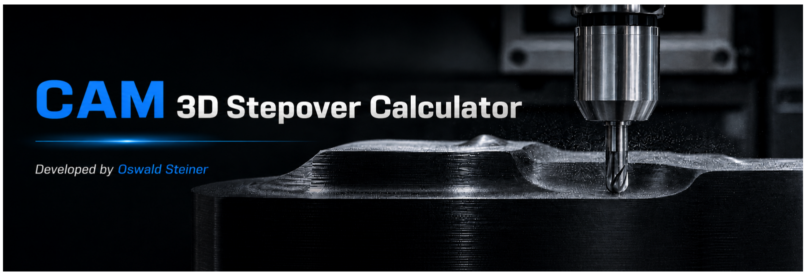

QUICK GUIDE
VERSION 3.2.1
WINDOWS APPLICATION

CAM 3D Stepover Calculator

A practical guide to finishing, semi-finishing and machine-limited speeds and feeds.

Purpose and features

Calculate starting stepovers for ball-nose and corner-radius tools. The app provides finishing stock guidance, automatic material-based spindle speed and feed estimates, and German/English switching for controls, results and the compact banner. The responsive layout rearranges panels; scroll vertically to reach all content in smaller windows.

Getting started

1. Launch **CAM_3D_Zustellrechner_v3_2_1.exe** and select **English**.
2. Choose the tool type and enter its diameter. For a corner-radius tool, enter the corner radius. Set the finishing Ra target; tool stickout is optional.
3. Select the material, operation and calculation mode: Practical, Conservative or Theoretical. For semi-finishing, adjust the default **+20%** stepover increase if needed.
4. Enter **Max. spindle [rpm]**, then click **Calculate**. Use **Copy result** to copy the readout. **Reset** restores tool defaults while retaining the machine limit.

Spindle speed and feed: a 10% machine reserve

Material and nominal tool diameter select an automatic starting profile. No manual cutting-speed or feed-per-tooth entry is required. The machine setting establishes a ceiling; it does not increase a lower calculated speed.

Applied rpm = min(profile rpm, 0.90 x maximum machine rpm)
Applied feed = profile feed x applied rpm / profile rpm

Example: A machine maximum of **20,000 rpm** permits at most **18,000 rpm**. If a profile gives 24,000 rpm and 1,200 mm/min, the applied values are 18,000 rpm and 900 mm/min. A profile already below 18,000 rpm remains unchanged. Leave the field blank to view uncapped profile values only.

Reading the results

3D stepover / ae shows the active operation. Semi-finishing defaults to 20% above the finishing reference (for example, 0.20 mm becomes 0.24 mm), subject to geometric limits. The Ra target applies to the subsequent finishing pass, not to semi-finishing. **Finish stock** is shown where a material table is available; no stock value is supplied for glass. The speed/feed panel shows both the profile values and the machine-adjusted values.

Materials and application notes

Materials: Graphite, aluminium, copper, steel (42CrMo4), stainless steel, hardened steel, plastic and glass.

Profiles are estimated starting values, not validated ideal cutting data. They assume suitable two-flute carbide finishing tools (diamond-coated for graphite); glass assumes a rounded diamond abrasive tool and feed per revolution. Nominal diameter is used, not the effective ball-nose contact diameter. Glass profiles and the geometric Ra model are not validated for a specific glass type. Check tooling, remaining stock, stability and cooling, and verify the result with a test cut or measurement.